

Reference excerpt 01-041

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the expression for the pressure gradient: $p_z = 1/c \cdot \partial \text{CSA} / \partial t$ (7) Flow velocity calculation $w(t, z)$ The function $w(t, z)$ is calculated by inserting into the Eq.(1) the expression for the pressure gradient found in Eq. (7). The mathematical details are given in [Sisini et al.(2015b)]. 3 Figure 3: Calculated velocity trace together with the experimental one obtained by the Doppler examination. RESULTS EXAMPLE Pressure waves velocity calculation The measured delay (t_{aQP}) was 0.14 s and the distance l_G was 23 cm. As a consequence the velocity c was 164 cm/s. Compliance calculation The minimum value of the CSA (CSA_x) during the cardiac cycle was 0.2 cm² and the compliance for unit of length was $9.8 \cdot 10^{-3}$ cm²/mmHg. Flow velocity calculation $w(t, z)$ The velocity was calculated following Eq. (1) and its trace is shown in Fig. 3 together with the experimental trace obtained by the Doppler examination. The two traces in agreement, nevertheless, since the velocity $w(t, G + l)$ was calculated up to an additive constant, only the wave amplitude has physical meaning whereas its shift along the velocity axis is not significant. ACKNOWLEDGEMENT The author wants to thank Eleuterio Toro for his critical comments to the paper and Giacomo Gadda, Valentina Tavoni and Valentina Sisini to have kindly reviewed the manuscript. REFERENCES [Applefeld(1990)] Applefeld MM. The Jugular Venous Pressure and Pulse Contour. In: Source Clinical Methods: The History, Physical, and Laboratory Examinations. Boston: Butterworths, 1990. chapter 19. [Beulen et al.(2011)] Beulen, B. et al. Toward noninvasive